Design and Finite Element Analysis (Static and Modal) of Formula Student Electric Vehicle Chassis

1.1 Introduction

University	Tribhuwan University
Semester	Sixth
College	IOE, Thapathali Campus
Subject	Team Shireto
Location	Kathmandu, Nepal
Project Duration	April to July 2023
Supervisor Name	Saugat Pandey
Degree	Bachelor of Mechanical Engineering

1.2 Background

1.2.1 Overview

I was with Team shireto which is a student-based group for research and development of electric vehicles. For this specific project we were planning to research and develop a formula student like car for the formula bharat competition that was about to held in 2024 in India. Being the design and FEA lead in the project, I worked with a team of 4 people for the complete design and analysis of the vehicle model. I was assigned with the work of chassis design and analysis along with material selection. Initially, starting with the conceptual designs as well as the regulations that were provided by Formula Bharat. The design has to have all the necessary strength capabilities to handle all the necessary testing scenarios. SolidWorks was utilized for 3D modeling for chassis design. ANSYS is applied to conduct structural analysis. It examines stress concentration, force distribution and failure points across all the different joints and loading areas.

The task of material selection was conducted to study different materials that would provide the strength and endurance along with being easily available and easy to work with. The ideas regarding chassis design were explored. AISI 4130 was selected as the material for the chassis frame. The 3D model was then created in SOLIDWORKS, where geometry was created with the planes and measurements for different sections as per the guidelines, followed by that the use of lines, points and planes. The model was brought to ANSYS for FEA simulation. Simulations were carried out for torsonal testing, static vertical bending, side impact test, and frontal impact test.

1.2.2 Objectives

The task motive was to design a chassis that can handle all the necessary forces during the working conditions of the vehicle.

Secondary goals were:

- To improve the strength of the chassis by selecting the most effective materials for improving the strength of the chassis.
- To improve the structural integrity of the chassis by analysis of the structure using FEA.

1.2.3 Nature of Work

I researched for related chassis designs from various literary sources. I selected materials and testing conditions based on engineering needs and performed simulations under various test conditions. I designed the 3D model in SOLIDWORKS and performed FEA in ANSYS Mechanical. I collaborated with my team members in design verification and result analysis to ensure that all activities were harmonized. I managed the workflow, set deadlines, and resolved technical problems to ensure consistent progress. I allocated tasks effectively, held regular meetings to track progress, and achieved on-time completion of the design and simulation stages.

1.2.4 Hierarchy

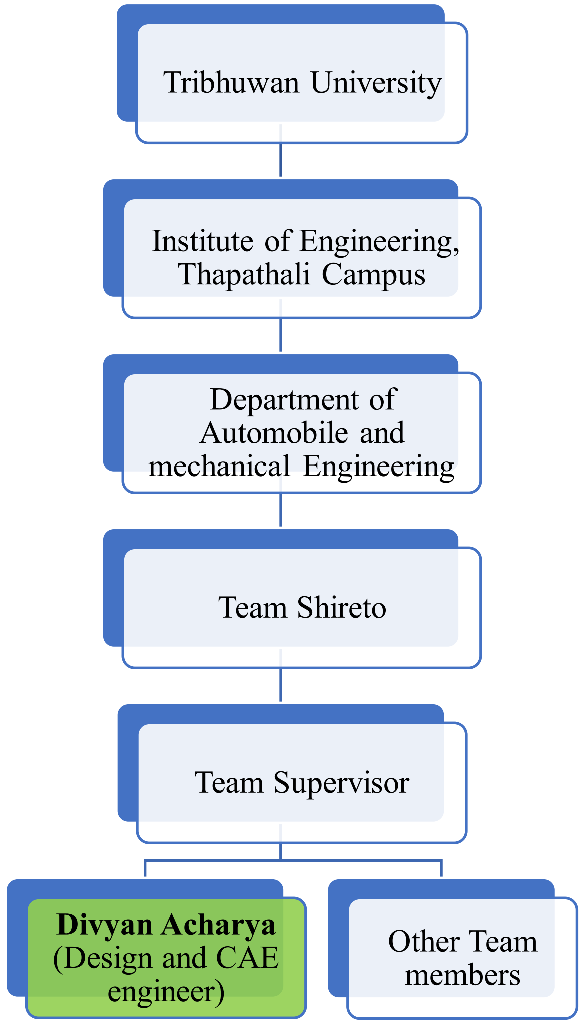

Figure1Hierarchy

1.2.5 Duties

- To research the project requirements for performance evaluation by reviewing literary sources.
- To select appropriate software, material, and define chassis dimensions for performance efficiency.
- To design the chassis model in SOLIDWORKS using planes, line sketches, points and symmetry tools.
- To perform FEA analysis in ANSYS Mechanical with various tests of a chassis.

- To evaluate result parameters across varying tests for performance assessment.

1.3. Personal Engineering Activities

1.3.1

I learned the fundamentals of chassis design and understood their strength capabilities with truss-based design. I got ideas from research papers and journals related to different types of chassis. I looked at various types of chassis to study differences in strength, weight and performance. I learned about how tubular spaceframe chassis provide the most efficient design for overall performance of the vehicle. I acquired ideas related to SolidWorks to make 3D model of chassis accurately. I studied the simulation process in ANSYS Mechanical for the study of strength and performance values. I studied the mesh refinement boundary conditions setup. I gained insights on balancing computational accuracy and mesh complexity for improved results.

1.3.2

I nominate AISI 4130 as the chassis material due to its high modulus of elasticity of 200GPa. I took AISI 4130 because of its high tensile strength (ultimate) of 560 MPa and excellent vibration damping properties. I choose tubular spaceframe as a chassis structure because of balanced strength, easy fabrication and lightweight design. I opted to use various links to comfort chassis and vehicle behavior. The designed chassis had an overall length of 2270 mm and a width of 604.844mm. I opted a 25.4mm diameter hollow pipe with 1.6 mm thickness for the design of tubular spaceframe chassis. The front bulkhead, side impact structure and base supporting structure were all designed with consideration of impact absorption and load management. I chose SOLIDWORKS weldments feature to model the chassis with precision. I used ANSYS for FEA analysis to visualize stress concerntrations during different testing conditions.

1.3.3

I began the initial sketch with different planes at different sections of the chassis such as front bulkhead, front roll hoop, main roll hoop, and rear bulkhead. Followed by that, I started with different points and lines using 3D sketch from weldments feature of Solidworks. I used the 3D sketch tool to lines for further weldment application and carefully gave dimentions accordingly. I mirrored the chassis using the “Mirror” feature to maintain symmetry along the longitudinal plane of the chassis. I created multiple front support structure, sideimpact support structure and bracing support areas to keepup the overall structural intergrety of the design. I then used intersection feature of the weldments to join different areas of multiple joints to create a smooth finished joint structure. Finally, I saved the complete chassis model in STEP format for further simulation.

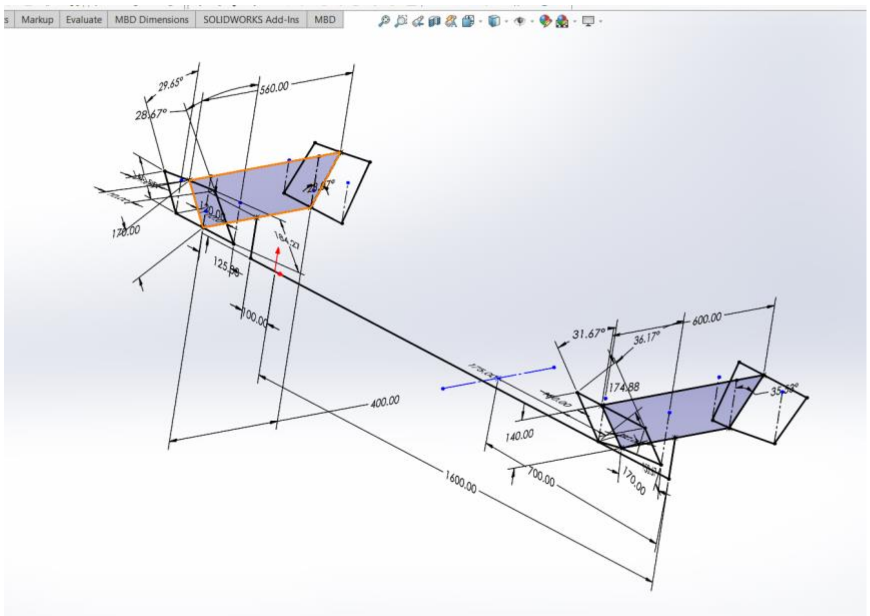

Figure2: Initial sketch with suspension hard points

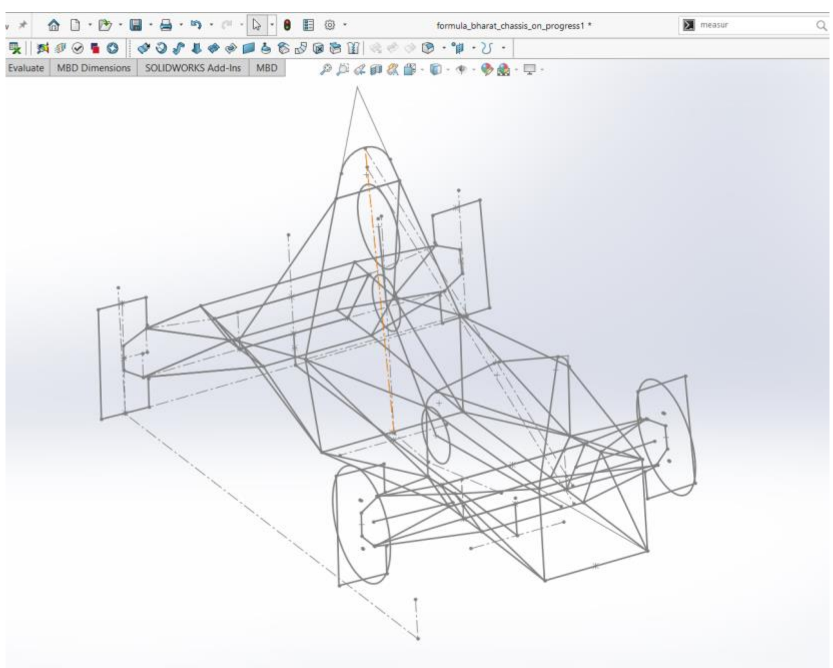

Figure3 : Fully defined 3D sketch with frontal and rear control arms

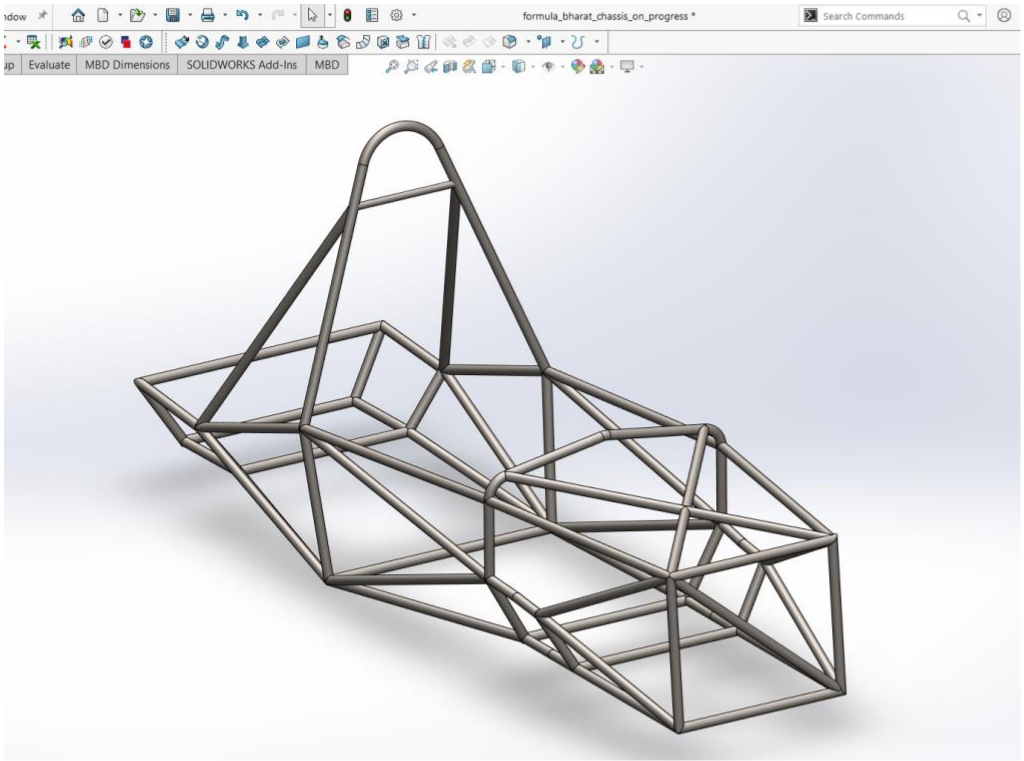

Figure4: Final 3D design

1.3.4

I started the FEA analysis of the chassis in ANSYS by first importing the STEP model into ANSYS Workbench. I used spaceclaim to clean the geometry and extract the mid-surface of the design. I performed a detail geometry cleanup with all necessary preparation for the joint sections of the chassis for precise analysis. I then opened the Meshing module in Ansys mechanical and generated a fine mesh with triangular elements, ensuring accurate results with 5mm mesh size and 1,09,813 nodes and 2,17,010 elements. The estimated weight of the car was about 265kg, including major driver's weight of 65kg and combined weight of battery pack and motor with 120kg.

I then started with the setup of different tests such as static vertical bending test with loads such as motor, battery, driver, and other heavy parts, side impact test with impacting force of 7798.9N simulating a 3G acceleration during turning, along with that a frontal impact test with impacting force of 12,998.25N simulating a 5G acceleration condition. Finally, there is a torsonal bending test with a couple of 3000N acting in the front suspension. I choose the solver of static structural from Ansys for the analysis.

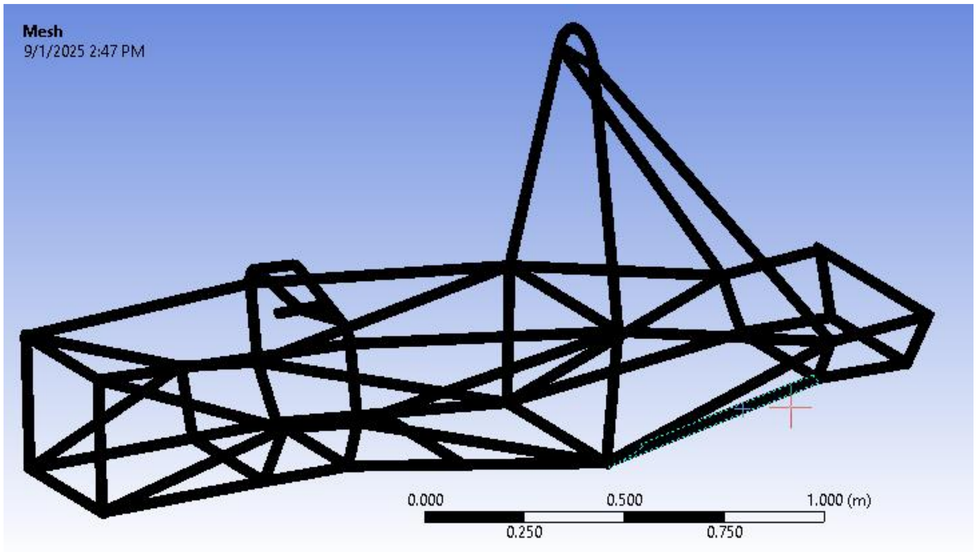

Figure4: Meshed model of the chassis

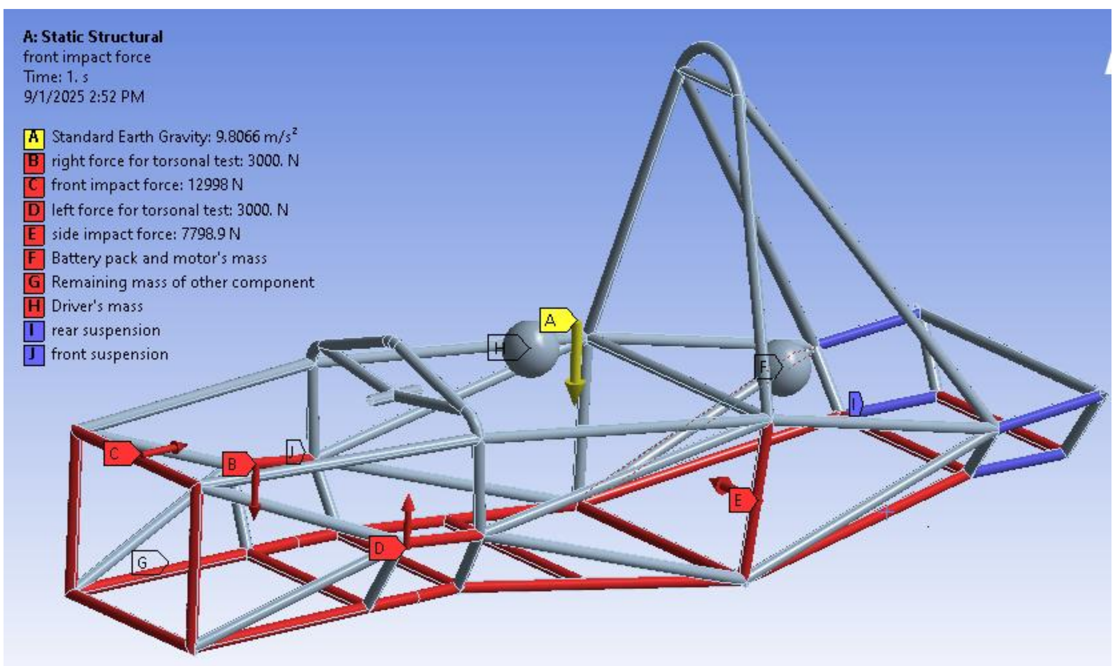

Figure5: Boundary conditions for all the necessary tests

1.3.5

I performed all the analysis as per the boundary conditions and assumed forces. The during all the four different tests, total deformation and equivalent (von-mises) stress were observed and analyzed. The total deformation of all the tests came out within the tolerance zone and the equivalent (von-mises) stress was also under the ultimate tensile strength of the material making the structure away from the failure.

Type of test	Deformation (mm)	Equivalent (von-mises) stress (Mpa)	FOS
Side impact test	5.24	391.71	1.42
Front impact test	1.04	267.24	2.09
Torsonal bending test	13.17	490.55	1.14
Static bending test	4.94	236.23	2.37

Result

For side impact test with 7798.9N force

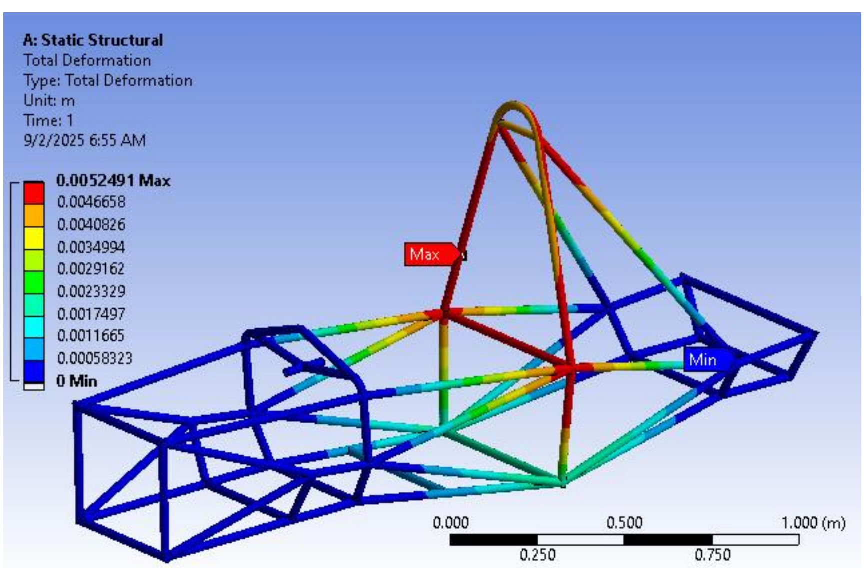

Figure6: Total deformation

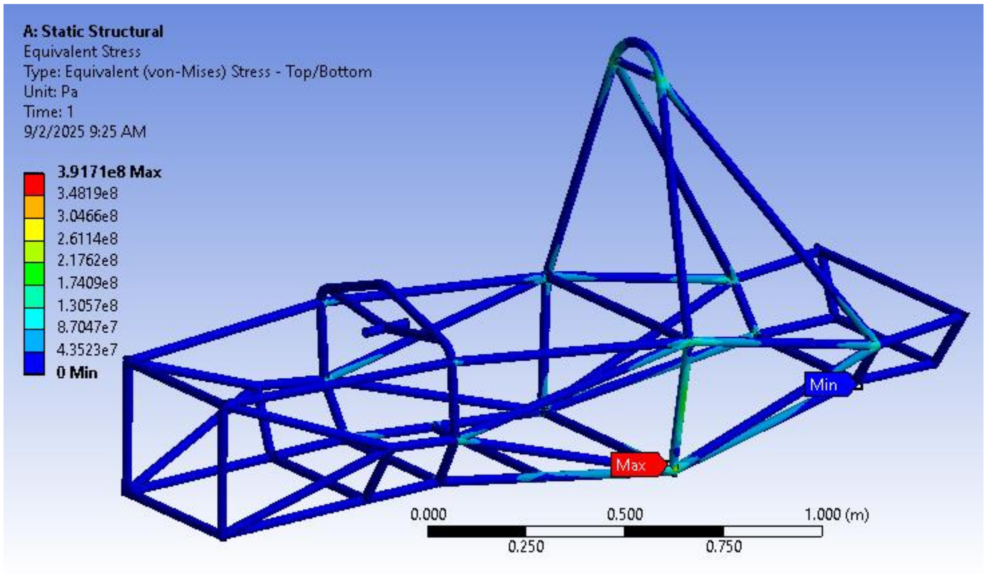

Figure7: Equivalent (Von-mises) stress

For front impact test with 12998.25N force

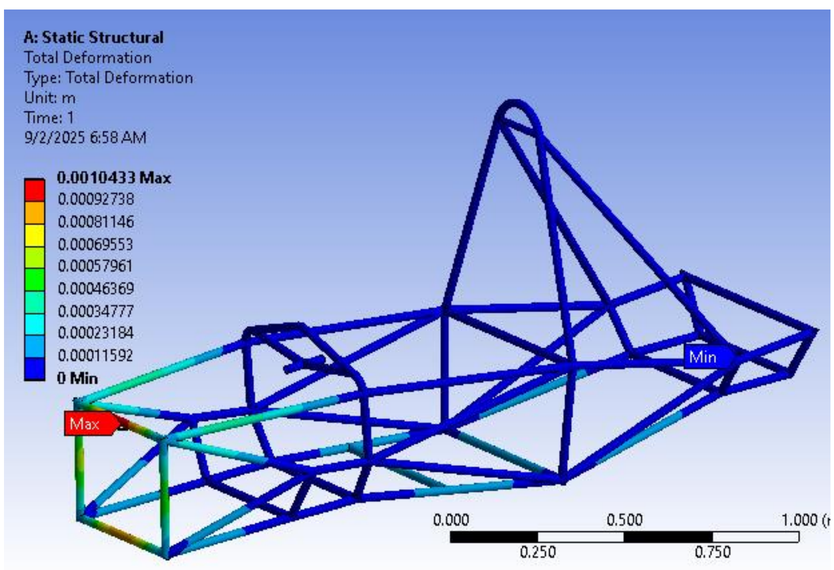

Figure8: Total deformation

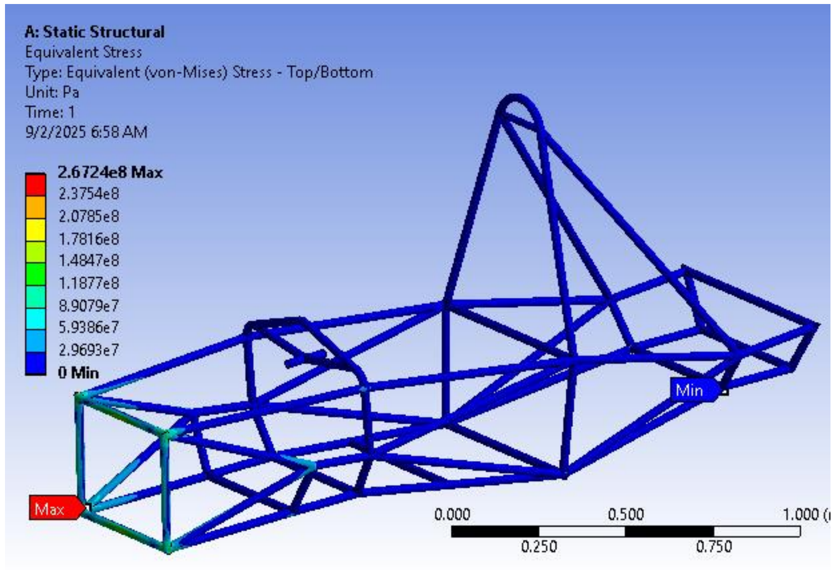

Figure8: Equivalent (Von-mises) stress

For torsonal bending test with 3000N couple in front suspension points

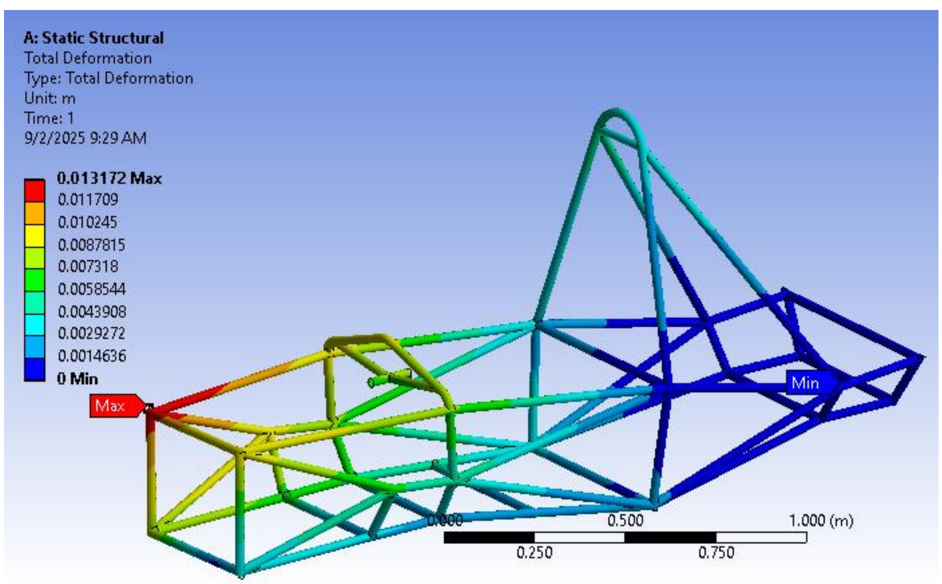

Figure9: Total deformation

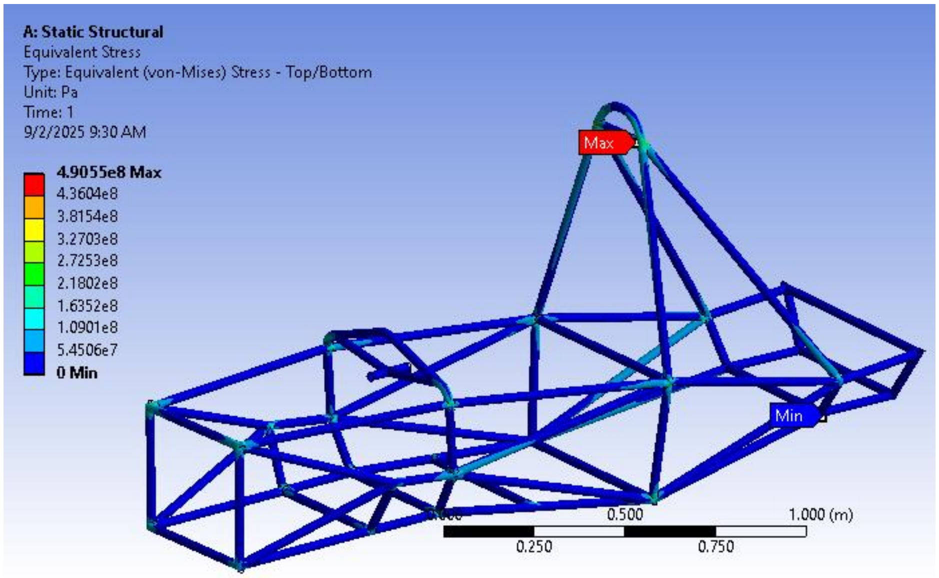

Figure10: Equivalent (Von-mises) stress

For static bending test with overall vehicle weight of 265kg

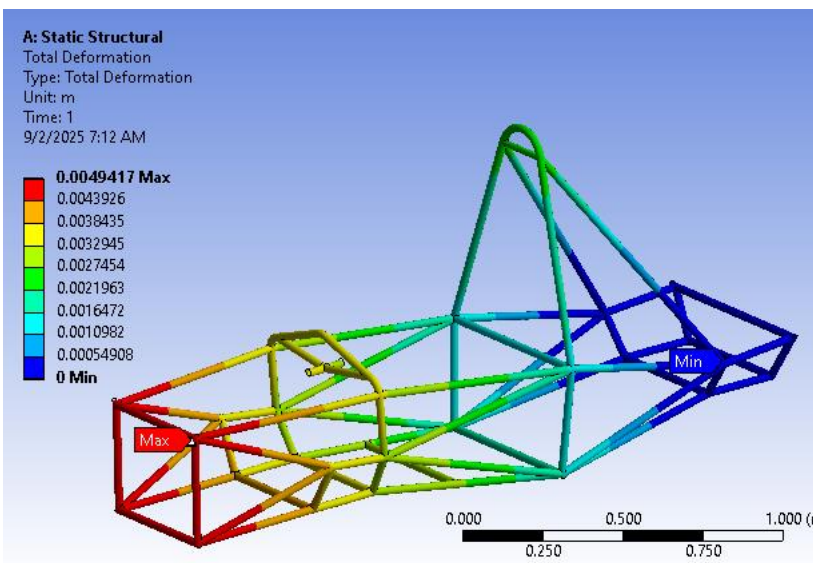

Figure11: Total deformation

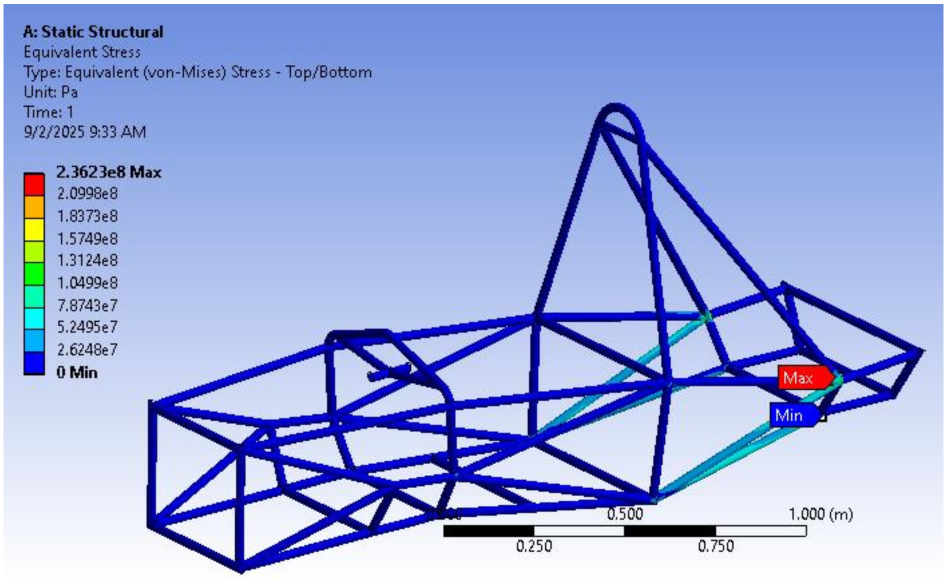

Figure12: Equivalent (Von-mises) stress

Prototype with PVC pipe of same diameter

1.4 Problem and Solution

I faced an issue at the FEA simulation when the stress and deformation values were unrealistic. I found that the reason was the due to improper load and boundary conditions assignment. I also faced problem while meshing due to improper geometry cleanup in the joints of the pipe. I noted that the FEA results started to show precise estimated values after all the corrections. I went to my team manager and supervisor to explain the inaccuracies of the simulation. I recommended several changes to increase the structural integrity of the model without significantly increasing its weight. As a result chassis became stronger, with reduced deformation and improved stress resistance.

1.5 Creative Works

I made a 3D model of chassis in SolidWorks with precise dimensions. I performed FEA simulations of the chassis for various tests in Ansys. I solved the results inaccuracy by making changes in boundary conditions. I made few changes for addition of more strength to the chassis.

1.6 Team Management

I was actively involved with the team throughout project. I managed and double-checked dimensional accuracy, estimated weight, materials and boundary conditions setup for accurate results. I cooperated with my team members and shared ideas and suggestions for more accurate testing of the chassis. I organized review meetings at regular intervals to check progress and maintain timely closure of deadlines. I organized coordination between documentation and modeling groups for uniformity in each step. I created a common shared folder to provide easy access to files as well as version control. I resolved disputes by listening to problems and making decisions by reason and technical rationale. I created a team culture of learning and innovation. I made sure that the end report included everyone's contribution. I successfully led the team from the research phase to completion simulation, and result analysis.

1.7 summary

1.7.1

The project was attained by successfully designing and analyzing a chassis at testing conditions. The structural design of the chassis was investigated in the beginning. SolidWorks was employed to create the 3D chassis model with the incorporation of features like 3D sketch, weldments, interference, mirror operations, and joint techniques. The developed model was brought into ANSYS, where mesh generation, material assignment, FEA boundary conditions were established. Four different types of testing was carried out with respective forces and constrains to find out the behavior of chassis in different test conditions. Resulting parameters were tracked for various tests. The overall strength of the chassis was enough to resist estimated forces in different situations without causing any failure of the material. Factor of safety (FOS) for all the tests were above 1 indicating high strength without failure. The chassis design functioned best when it was tested in static bending test with the overall estimated weight of the vehicle. It achieved the aims through proper design modeling and extensive simulation analysis that validated strong performance improvements with a design with high structural integrity.

1.8.2

I developed strong modeling skills in SOLIDWORKS and learned to simulate different tests in ANSYS. I improved my FEA analysis abilities and gained hands-on experience the design and simulation modeling process along with development of prototype. I enhanced my teamwork, problem-solving, and technical communication skills.